

Factors Relating to the Production of a Radiograph

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Radiation Quantity and Quality

Radiation Quantity is the **number** of x-ray photons in the useful beam. The factors affecting x-ray quantity are:

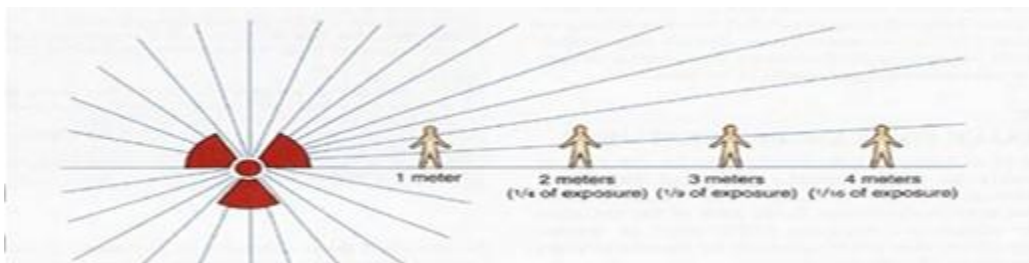
- **mAs:** X-ray quantity is directly proportional to milliamperere-seconds.
- **kVp:** X-ray quantity is directly proportional to kilovolt peak.
- **Distance:** X-ray quantity varies inversely with distance.
- **Filtration:** X-ray quantity is reduced by filtration, which absorbs the low-energy photons of the beam.

Radiation Quality is the penetrating power of the x-ray beam, which is quantified by HVL (Half-Value Layer). The factors affecting x-ray quality are:

- **kVp:** X-ray penetrability is increased as kVp is increased.
 - **Filtration:** X-ray penetrability is increased when filters are added to the beam.
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Inverse Square Law

The law states that the intensity of radiation is inversely proportional to the square of the distance measured from the source to the point of measurement.



Factors in Radiograph Production

Factors relating to the production of a radiograph can be divided into three main categories:

- **A.** Factors related to the radiation beam.
- **B.** Factors related to the object.

- C. Factors related to the X-ray film.
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A. Factors Related to the Radiation Beam

1. **Exposure time:** The interval during which X-rays are being produced. Exposure time is directly related to the total photon production; thus, increasing exposure time causes an increase in the quantity of X-radiation. That's why exposure time has a direct effect on film density.
 2. **Milliamperage (mA):** This is related to the amount of electricity passing through the filament circuit. It directly controls the rate of X-ray photon production and thus has a direct effect on film density.
 3. **Kilovoltage (kV):** This refers to the potential difference between the cathode and anode in the x-ray tube. The higher the kVp, the greater the potential difference and the greater the energy of the X-ray photons.
 4. **Tube-film distance:** This distance consists of the tube-object distance and the object-film distance.
 - It affects the intensity of radiation according to the inverse square law.
 - It affects the exposure time directly.
 - It affects the radiation dose. Decreasing the tube-film distance makes the X-ray beam more divergent behind the skin area, and more tissue is irradiated. Increasing the distance makes the beam less divergent and reduces the amount of tissue irradiated.
 5. **Focal spot size:** The focal spot, or the source of radiation, must be as small as possible to get the best image quality. Any movement in the head of the X-ray machine affects the focal spot size.
 6. **Collimation:** A collimator is used to control the size and shape of the beam. Collimation reduces the amount of tissue irradiated and minimizes the production of secondary radiation fog. **Fog** is unwanted film density (blackening) that reduces radiographic contrast.
 7. **Filtration:** The effect of filtration is the absorption of long-wavelength X-ray photons that have low penetrating power. The result is a "hardened" beam (more short-wavelength photons with high penetration power). Increased filtration decreases contrast and also decreases density because it absorbs some short-wavelength photons in addition to the long-wavelength ones.
 8. **Equipment efficiency:** Dental X-ray machines differ in construction and efficiency, so the quality and quantity of the X-ray beam vary from one machine to another.
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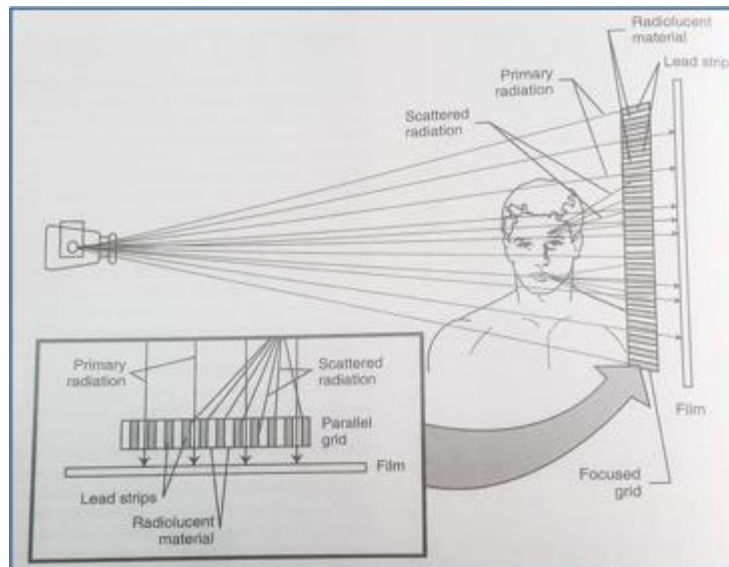
B. Factors Relating to the Object

1. **Thickness of the object:** A thick object requires more radiation to create a radiographic image. It is often advisable to increase kV or mA and/or exposure time to increase the number of X-ray photons.
2. **Density of the object:** Density refers to the weight per unit volume. In dental radiography, tooth enamel has the highest density of all body tissues. Increasing the

density of an object increases its ability to absorb X-radiation. Hard tissue like enamel absorbs a great amount of radiation compared to soft tissue like the pulp.

C. Factors Relating to the X-ray Film

1. **Reduction of secondary radiation:** Secondary radiation (scattered, stray, leakage) is undesirable because it reaches all parts of the film and produces film fog. It can be minimized by:
 - Using as small a beam of radiation as possible.
 - Proper collimation.
 - Using a sheet of lead foil behind the film in intraoral film packets.
 - Using a **grid** between the object and the film for extraoral films. A grid is composed of alternating strips of radiopaque (lead) and radiolucent (plastic) material. It transmits X-rays traveling in a straight line from the source to the film and absorbs scattered radiation.



2. **Film and film storage:** X-ray film must be stored in light-tight, lead-lined containers away from stray radiation, excessive temperature, or humidity. It should be used before its expiration date.
 3. **Intensifying screen:** This device converts the energy of the X-ray beam into visible light, which then exposes the film. Intensifying screens are used with extraoral films to reduce the patient's dose, as one X-ray photon can generate many light photons.
 4. **Film processing:** The latent image is formed when silver halide grains are exposed to X-rays. During development, only the exposed grains form the visible image. The unexposed grains are removed by the fixer to make the image permanent. Automatic processing is preferred because it is faster and results in better image quality.
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Ideal Radiographic Projection

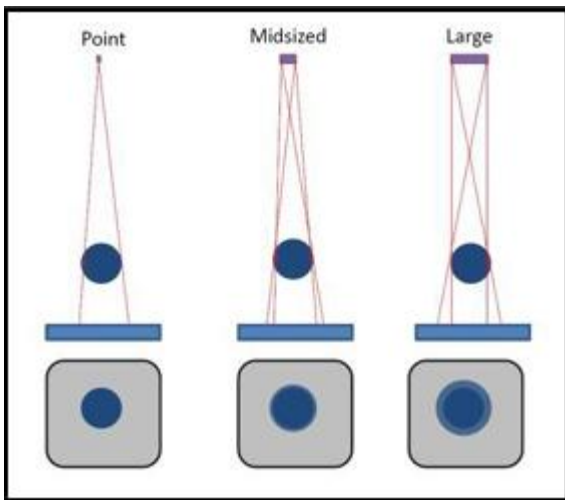
Image quality is a clinician's subjective judgment of a radiograph's overall appearance. An ideal radiograph demonstrates:

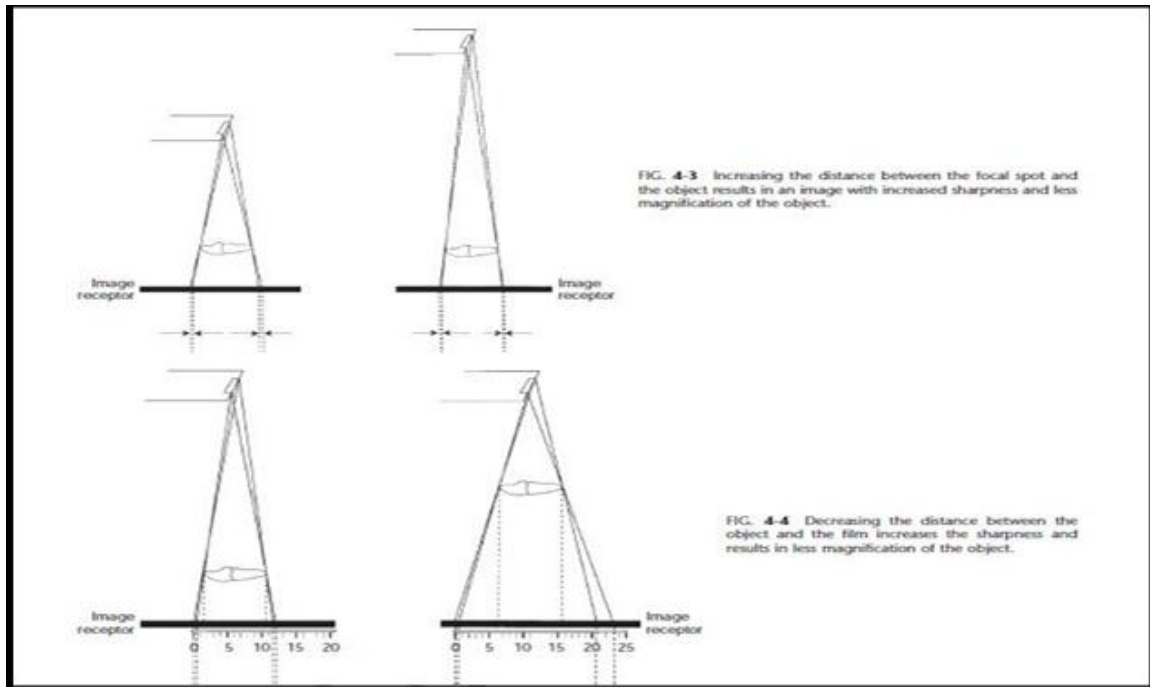
- An image that is **sharp**.
- An image that is **shaped like the object**.
- An image that is the **same size as the object**.

Key Terms:

- **Sharpness:** The ability of a radiograph to define an edge precisely.
- **Penumbra:** The area of partial shadow or unsharpness on the image. It is created by the size of the focal spot—the larger the spot, the greater the penumbra.
- **Umbra:** The area of total shadow.

[Image showing effect of focal spot size on penumbra]





Basic Principles of Projection Geometry

1. The source of radiation should be as **small** as possible.
2. The tube-to-object distance should be as **great** as possible.
3. The object-to-film distance should be as **small** as possible.
4. The film should be **parallel** to the object.
5. The central ray of the beam should be **perpendicular** to the film.

Radiographic Errors and Artifacts

Errors are classified into three categories:

- A. Technique and projection errors
- B. Exposure errors
- C. Processing errors

Common Errors:

1. **Cone cut:** A clear, unexposed area resulting from the X-ray beam not completely covering the film.
2. **Back side exposure:** The image has a pattern from the lead foil because the film was placed backward.
3. **Double exposure:** An excessively dense and blurred image from exposing the same film twice.
4. **Elongated image:** Occurs when the vertical angulation of the X-ray tube is too shallow.

5. **Shortened image:** Occurs when the vertical angulation is too steep.
6. **Overlapping:** Adjacent structures overlap when the horizontal angulation is incorrect.
7. **Blurred film:** Caused by excessive bending of the film during placement.
8. **Pale film:** Caused by underexposure or underdevelopment.
9. **Dark film:** Caused by overexposure or overdevelopment.
10. **Completely clear film:** Occurs if the film receives no radiation or is placed in the fixer before the developer.
11. **Undeveloped area:** A clear area caused by the film not being fully immersed in the developer.
12. **Scratched film:** The soft emulsion is scratched by rough handling during manual processing.
13. **Developer spot:** Dark spots caused by developer solution spilling on the film before processing.
14. **Fixer spot:** White or light spots caused by fixer solution spilling on the film before processing.
15. **Yellow or brown stain:** Discoloration due to contaminated solutions or insufficient rinsing.



Cone cut



back side exposure



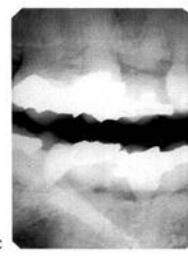
double exposure



Elongation



shortening



overlapping



Blurred image



Pale film



dark film



undeveloped area



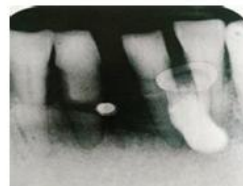
scratched film



developer spot



Developer spot (occlusal film)



fixer spot



complete clea



yellow or brown discoloration